

Practical No : 1

Terminal 1 :

```
javac *.java
rmiregistry
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment... × bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment..
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ javac *.java
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ rmiregistry
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System::setSecurityManager has been called by sun.rmi.registry.RegistryImpl
WARNING: Please consider reporting this to the maintainers of sun.rmi.registry.RegistryImpl
WARNING: System::setSecurityManager will be removed in a future release
```

Terminal 2 :

```
javac Server.java
java Server
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment... × bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment..
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ javac server
error: Class names, 'server', are only accepted if annotation processing is explicitly requested
1 error
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ javac Server.java
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Server
Server Started....
^Cbhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Server
Server Started....
```

Terminal 3 :

```
java Client
Enter 1st String : abc
Enter 2nd String : xyz
then Output
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment... × bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment..
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Client
Enter First String: Bhushan
Enter Second String: Ahire
----- Results -----
Strings After Joining Is: BhushanAhire
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ ^C
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Client
```

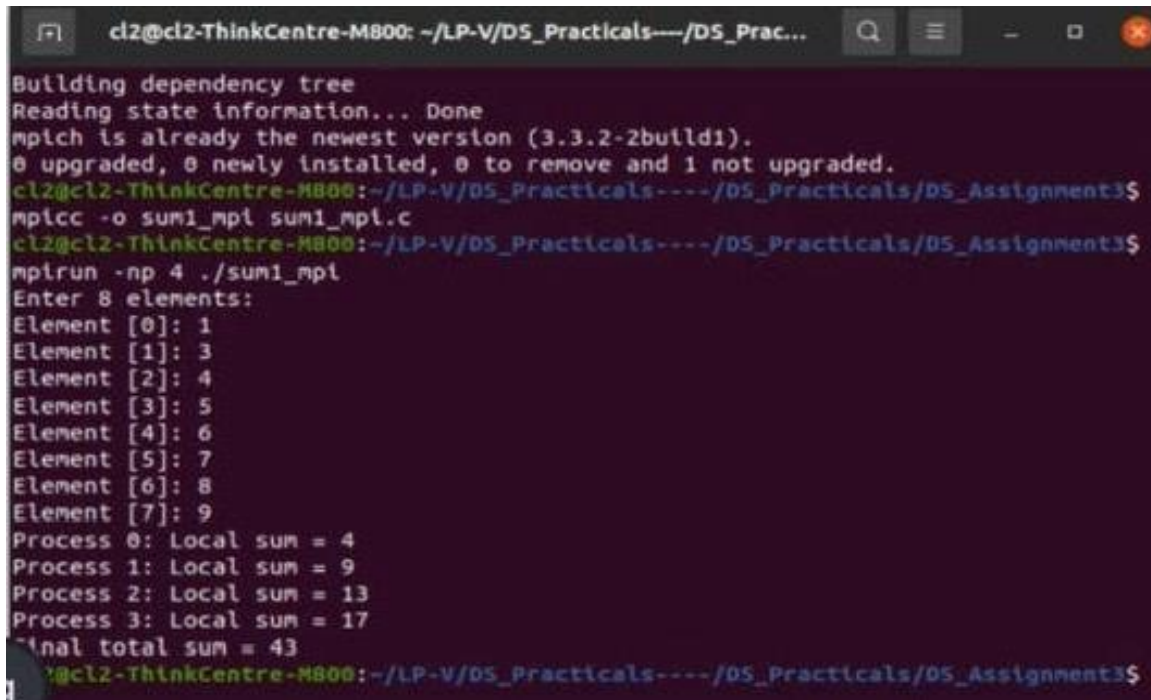
Practical No : 3

installation :

```
sudo apt update  
sudo apt install mpich
```

Terminal 1 :

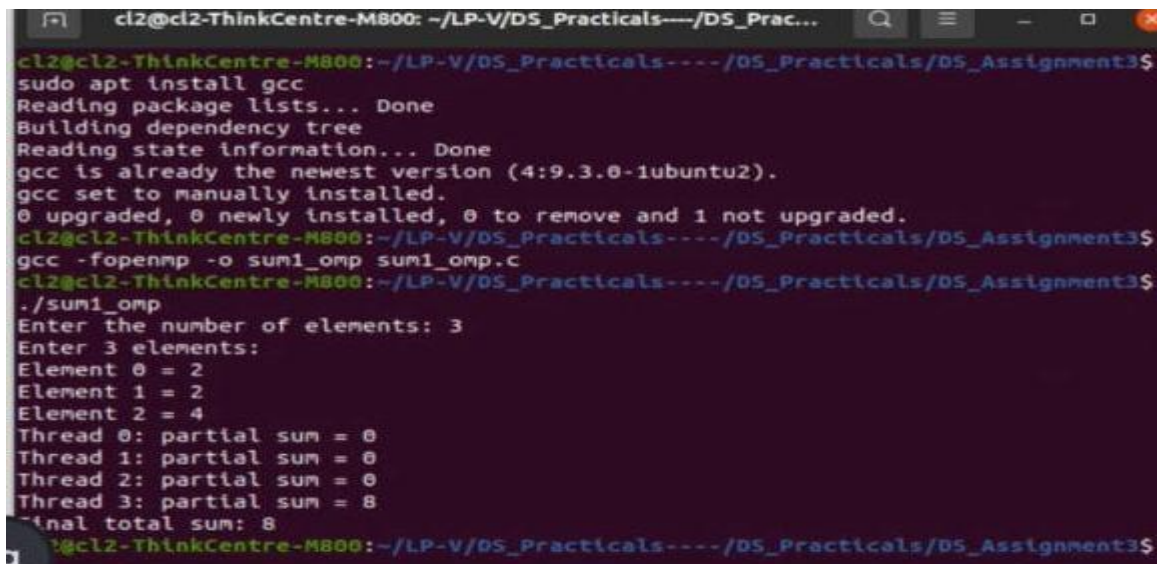
```
mpicc -o sum1_mpi sum1_mpi.c  
mpirun -np 4 ./sum1_mpi
```

A terminal window titled 'cl2@cl2-ThinkCentre-M800: ~/LP-V/DS_Practicals---/DS_Prac...' showing the execution of an MPI program. The user runs 'mpicc -o sum1_mpi sum1_mpi.c' and then 'mpirun -np 4 ./sum1_mpi'. The program prompts for 8 elements, which are entered as 1, 3, 4, 5, 6, 7, 8, and 9. It then displays the local sum for each of the 4 processes (0, 1, 2, 3) and the final total sum of 43.

```
cl2@cl2-ThinkCentre-M800: ~/LP-V/DS_Practicals---/DS_Prac...  
Building dependency tree  
Reading state information... Done  
mpich is already the newest version (3.3.2-2build1).  
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
mpicc -o sum1_mpi sum1_mpi.c  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
mpirun -np 4 ./sum1_mpi  
Enter 8 elements:  
Element [0]: 1  
Element [1]: 3  
Element [2]: 4  
Element [3]: 5  
Element [4]: 6  
Element [5]: 7  
Element [6]: 8  
Element [7]: 9  
Process 0: Local sum = 4  
Process 1: Local sum = 9  
Process 2: Local sum = 13  
Process 3: Local sum = 17  
Final total sum = 43  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$
```

Terminal 2 :-

```
Sudo apt install gcc  
gcc -fopenmp -o sum1_omp sum1_omp.c  
./sum1_omp
```

A terminal window titled 'cl2@cl2-ThinkCentre-M800: ~/LP-V/DS_Practicals---/DS_Prac...' showing the installation of gcc and the execution of an OpenMP program. The user runs 'sudo apt install gcc', then 'gcc -fopenmp -o sum1_omp sum1_omp.c', and finally './sum1_omp'. The program prompts for the number of elements (3) and their values (2, 2, 4). It then displays the partial sum for each of the 4 threads (0, 1, 2, 3) and the final total sum of 8.

```
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
sudo apt install gcc  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
gcc is already the newest version (4:9.3.0-1ubuntu2).  
gcc set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
gcc -fopenmp -o sum1_omp sum1_omp.c  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
./sum1_omp  
Enter the number of elements: 3  
Enter 3 elements:  
Element 0 = 2  
Element 1 = 2  
Element 2 = 4  
Thread 0: partial sum = 0  
Thread 1: partial sum = 0  
Thread 2: partial sum = 0  
Thread 3: partial sum = 8  
Final total sum: 8  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$
```

Practical No : 4

installation :

sudo apt install python-is-python3

Terminal 1 :

python3 server.py

```
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment4$ python3 server.py
Socket at master node created successfully

Clock server started...

Starting to make connections...

Starting synchronization parallelly...

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.
```

Terminal 2 :

python3 client.py

```
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment4$ python3 client.py
Starting to receive time from server

Starting to receiving synchronized time from server

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:13.316274

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:18.317180

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:23.318605

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:28.319290

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:33.319888

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:38.320849

Recent time sent successfully
Synchronized time at the client is: 2026-05-02 12:35:43.321768

Recent time sent successfully
```

Practical No:5

Practical No : 5

Terminal 1:

```
javac tokenring.java
java tokenring
```

```
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment5$ javac tokenring.java
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment5$ java tokenring
Enter the num of nodes:
5
 0 1 2 3 4 0
Enter sender:
1
Enter receiver:
4
Enter Data:
45
Token passing: 0-> 1
Sender 1 sending data: 45
data 45 forwarded by 2
data 45 forwarded by 3
Receiver 4 received data: 45

Do you want to send again?? enter 1 for Yes and 0 for No : 1
Enter sender:
2
Enter receiver:
2
Enter Data:
43
Token passing: 1-> 2
Sender 2 sending data: 43
data 43 forwarded by 3
data 43 forwarded by 4
data 43 forwarded by 0
data 43 forwarded by 1
Receiver 2 received data: 43

Do you want to send again?? enter 1 for Yes and 0 for No : 0
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment5$
```

Practical No : 6

Terminal 1 :

```
javac Bully.java
java Bully
```

```

bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignments$ javac Bully.java
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignments$ java Bully
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 1
Enter the number of processes:- 4
Creating processes..
P1 created
P2 created
P3 created
P4 created
Process P4 is the coordinator
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 2
P1 is up
P2 is up
P3 is up
P4 is up
Process P4 is the coordinator
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 3
Enter the process number to up:- 3
Process 3 is already up.
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm

```

Terminal No : 2

```
javac Ring.java
java Ring
```

```

bush@bush-HP-EliteBook-840-G3: ~/Downloads/DS_Practicals/DS_Assignment6
bush@bush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment6$ javac Ring.java
bush@bush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment6$ java Ring
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 1
Enter the total number of processes:- 4
P1 created.
P2 created.
P3 created.
P4 created.
P4 is the coordinator
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 2
P1 is up.
P2 is up.
P3 is up.
P4 is up.
P4 is the coordinator
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 3
Enter the process to up:- 2
Process P2 is already up.
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process

```